

ONAHAMA, Japan

WMO: 475980

Lat: 36.9475N

Lon: 140.9033E

Elev: 17

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.9	29.7	10.7	9.5	36.6	13.0	10.6	36.7	22.3	41.9	20.3	41.7	4.9	340	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	84.6	75.8	82.6	75.4	81.0	74.6	77.8	81.9	76.8	80.9	75.9	79.7	7.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	138.2	80.2	75.5	133.7	79.5	74.6	129.7	78.8	41.3	82.3	40.4	81.1	39.5	80.0	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.9	16.4	14.3	DB	24.2	91.5	1.7	2.8	23.0	93.5	22.0	95.2	21.0	96.8	19.7	98.8
			WB	21.6	79.0	1.4	1.2	20.6	79.9	19.7	80.6	18.9	81.3	17.9	82.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	39.3	39.7	44.6	52.8	60.5	66.6	73.3	76.7	71.8	62.8	52.7	43.9
	DBStd	13.63	3.84	4.13	4.96	5.25	4.43	4.13	3.93	3.65	4.84	4.95	5.11	4.87
	HDD50	1066	333	289	183	31	0	0	0	0	0	0	31	199
	HDD65	3813	797	707	633	366	148	29	1	0	5	103	370	654
	CDD50	3675	1	1	14	115	326	499	722	827	654	396	110	10
	CDD65	950	0	0	0	0	9	77	258	362	210	34	0	0
	CDH74	4453	0	0	0	2	17	120	1021	2316	921	56	0	0
	CDH80	594	0	0	0	0	1	11	112	368	96	6	0	0
(14) Wind	WSAvg	6.3	7.2	7.2	7.1	7.0	6.2	5.2	5.1	5.3	5.8	6.0	6.1	6.8
(15) Precipitation	PrecAvg	55.5	2.2	2.2	3.7	4.7	5.6	5.6	6.4	5.1	7.4	6.9	3.5	2.1
	PrecMax	76.2	6.1	4.6	6.5	9.0	10.4	11.3	13.0	13.5	19.1	17.9	8.3	5.5
	PrecMin	36.8	0.2	0.4	0.9	1.0	1.9	1.5	1.2	0.5	0.6	1.5	0.1	0.0
	PrecStd	7.8	1.5	1.3	1.5	2.4	2.0	2.4	2.8	3.4	4.1	4.4	2.1	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.5	57.5	63.3	70.0	76.3	81.2	86.2	88.6	85.4	79.0	69.1	63.0
		MCWB	48.4	48.7	53.7	56.7	61.4	67.5	75.0	76.2	74.7	67.4	60.0	56.2
	2%	DB	52.8	53.8	59.3	65.8	71.8	77.1	82.8	85.4	82.5	75.1	66.3	59.0
		MCWB	44.5	46.4	51.4	56.7	61.9	68.9	74.9	76.6	74.7	67.2	58.4	51.0
	5%	DB	50.1	51.1	56.7	63.4	69.3	74.7	80.7	83.6	80.6	72.8	64.4	56.1
		MCWB	42.2	43.8	49.6	55.7	61.5	68.1	74.4	76.4	73.8	65.5	57.8	48.7
	10%	DB	47.6	48.4	54.2	61.3	67.3	72.7	79.0	81.9	78.8	70.8	62.4	53.4
		MCWB	40.1	41.2	47.5	54.6	60.8	67.1	73.6	75.7	73.0	64.0	55.7	46.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	52.2	56.9	61.9	67.1	72.6	77.7	79.2	78.2	72.2	63.8	59.5
		MCDB	55.4	55.3	59.9	65.5	70.7	76.7	82.1	83.6	82.0	75.4	65.8	62.5
	2%	WB	46.3	47.9	53.6	59.4	64.9	71.0	76.4	78.3	76.7	69.8	61.3	53.0
		MCDB	50.9	52.3	57.8	62.9	68.5	74.8	80.5	82.5	80.3	72.9	64.0	56.7
	5%	WB	43.5	44.8	51.2	57.8	63.6	69.6	75.3	77.4	75.6	67.5	59.4	49.9
		MCDB	49.0	49.8	55.3	61.4	67.0	72.8	79.1	81.6	79.1	71.3	63.0	55.1
	10%	WB	40.6	42.0	48.5	56.1	62.2	68.2	74.3	76.7	74.4	65.4	56.9	46.9
		MCDB	46.6	47.4	53.1	60.0	65.7	71.3	78.0	80.7	77.9	69.7	61.6	52.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	14.3	14.3	13.5	11.5	9.6	8.9	9.1	10.3	12.6	14.9	15.0
		MCDBR	18.3	18.5	17.6	16.2	14.3	12.4	11.4	10.9	11.2	13.5	16.8	18.4
		MCWBR	12.9	13.8	13.3	11.0	8.6	6.6	5.6	5.1	6.2	8.8	12.1	13.5
	5% WB	MCDBR	17.6	17.6	16.5	13.6	11.8	10.5	10.2	9.7	9.2	11.4	14.6	16.8
		MCWBR	14.0	14.6	14.8	10.8	8.5	6.2	5.6	5.0	5.8	9.3	12.5	14.3
(40) Clear-Sky Solar Irradiance	taub		0.378	0.429	0.506	0.561	0.579	0.605	0.606	0.589	0.516	0.473	0.419	0.382
	taud		2.209	2.060	1.889	1.815	1.825	1.831	1.883	1.920	2.077	2.106	2.171	2.223
	Ebn at Noon		249	248	239	234	233	226	225	225	235	232	231	238
	Edh at Noon		34	45	58	66	67	66	62	59	48	43	36	32
(44) All-Sky Solar Radiation	RadAvg		795	998	1275	1508	1640	1530	1524	1513	1214	960	762	685
	RadStd		45	82	72	137	160	113	238	178	100	93	58	44

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page